THE UNIVERSITY OF ILLINOIS AT CHICAGO
Department of Chemical Engineering

Course: CHE 341. Chemical Process Control
Instructor: Dr. Michael Caracotsios
E-mail: mcaracot@uic.edu

Midterm #2B

Student Name: Kevin Grassmann

Due Date: Friday Apr-07 **Time:** 4:00pm

TAKE HOME EXAM RULES: PLEASE READ VERY CAREFRULLY

1. Do not share your code or solution with other students
2. Do not talk to other students or discuss the solution process
3. If you have questions you can only talk to your Professor

4. Prepare a solution report of the exam, typed in Microsoft Word
5. Clearly indicate all steps in the derivation of all formulas
6. State all references used during the solution process
7. Include in the report only what is asked and nothing more

8. Solution reports not typed will be returned to the student
9. If copying or sharing of the solution or programming code is found all reports will be nullified and an additional exam will take place in class.

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Problem 1

The process is represented in the Laplace domain by the following equation:

$$G_p = \frac{K_p e^{-\theta_p s}}{\tau_p s + 1}$$

Equation 1: FOTD System

Where, $\{K_p = 6.73,\ \theta_p = 8.47,\ \tau_p = 25.0\}$

The closed loop transfer function for the process in question is shown below:

$$1 + G_c G_p = 0$$

$$G_c = K_c$$

Equation 2: Closed Loop Transfer Function Characteristic Equation

Using the PID Simulation Excel program provided by Professor Michael Caracotsios, a proportional controller feedback loop was analyzed for the given system and through manual control, a critically damped Kc value was obtained.

$Kc_{true} = 0.122$

Two methods of obtaining the critically damped control gains were devised and compared to Kc_{true} in order to check the accuracy of the methods.

Method 1: Pade Approximation

The first devised method for calculating the critically damped control gain was using the Pade approximation. From the Pade approximation the first order time delay transfer function can be modeled as a second order critically damped transfer function.

$$e^{-s\theta} = \frac{2 - s\theta}{2 + s\theta}$$

Equation 3: Pade Approximation

$$\tau^2 s^2 + 2\tau s + 1 = 0$$

Equation 4: Critically Damped Second Order Function

Plugging equation 3 into equation 2 results in the new equation, equation 5.

$$1 + \frac{K_c K_p (2 - s\theta)}{(\tau_p s + 1)(2 + s\theta)} = 0$$

$$\frac{(\tau_p s + 1)(2 + s\theta) + K_c K_p (2 - s\theta)}{(\tau_p s + 1)(2 + s\theta)} = 0$$

$$\frac{\tau_p{}^2\theta_p s^2 + (2\tau_p + \theta_p - K_cK_p\theta_p)s + (2 + 2K_cK_p)}{(\tau_p s + 1)(2 + s\theta)} = 0$$

$$\tau_p{}^2\theta_p s^2 + (2\tau_p + \theta_p - K_cK_p\theta_p)s + (2 + 2K_cK_p) = 0$$

$$\frac{\tau_p{}^2\theta_p s^2}{(2 + 2K_cK_p)} + \frac{(2\tau_p + \theta_p - K_cK_p\theta_p)}{(2 + 2K_cK_p)}s + 1 = 0$$

Equation 5: Closed Loop Transfer Function with Pade Approximation

Comparing equation 5 and equation 4, an equation relating Kc to Kp, θp, and τp can be developed by relating the coefficients

$$\tau^2 = \frac{\tau_p{}^2\theta_p}{(2 + 2K_cK_p)}\,;\, 2\tau = \frac{(2\tau_p + \theta_p - K_cK_p\theta_p)}{(2 + 2K_cK_p)}$$

$$\boxed{\frac{\tau_p{}^2\theta_p}{(2 + 2K_cK_p)} = \left[\frac{(2\tau_p + \theta_p - K_cK_p\theta_p)}{2(2 + 2K_cK_p)}\right]^2}$$

Equation 6: Derived Kc Equation

Plugging equation 6 into GoalSeek results in a Kc value of 0.097 and percent error of 20.49%. Since Kc_{pade} is less than Kc_{true}, the system is overlapped, as shown in figure 1.1.

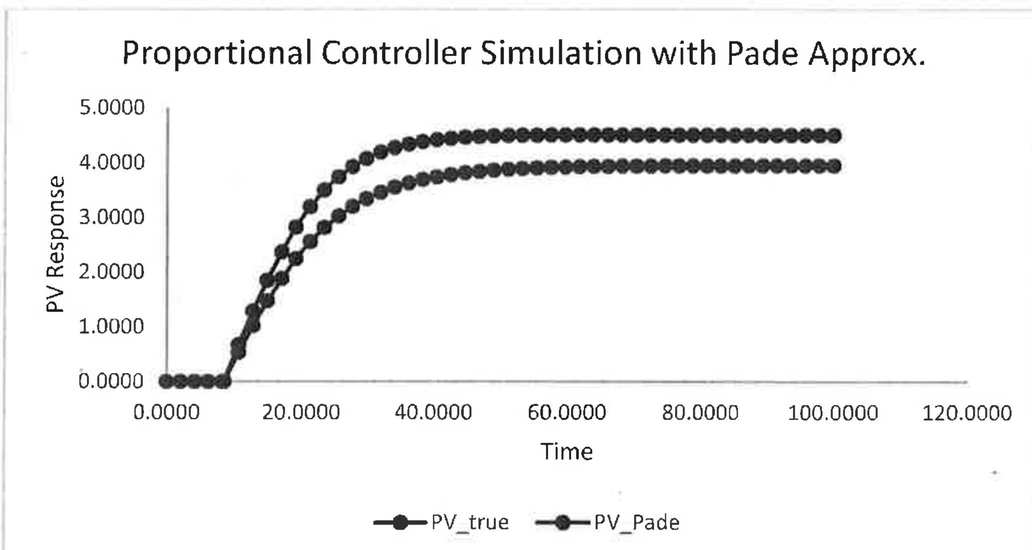

Figure 1.1: Comparison between Pade calculated gain controller and actual critical damped controller

Method 2: Second Order Taylor Expansion

A second order Taylor expansion of the time delay term can fit the characteristic transfer function to a critically damped second order equation, allowing for a method of calculating Kc.

$$e^{-s\theta} = 1 - \theta s + \frac{\theta^2 s^2}{2}$$

Equation 7: Second order Taylor expansion

Plugging in equation 7 into equation 2 gives the following equation:

$$\frac{(\tau_p s + 1 + K_c K_p - K_c K_p \theta s + \frac{K_c K_p \theta^2}{2} s^2)}{\tau_p s + 1} = 0$$

$$\frac{K_c K_p \theta^2}{2} s^2 + (\tau_p s - K_c K_p) s + (K_c K_p + 1) = 0$$

$$\frac{K_c K_p \theta^2}{2(K_c K_p + 1)} s^2 + \frac{(\tau_p s - K_c K_p)}{(K_c K_p + 1)} s + 1 = 0$$

$$\tau^2 = \frac{K_c K_p \theta^2}{2(K_c K_p + 1)}\ ; 2\tau = \frac{(\tau_p s - K_c K_p)}{(K_c K_p + 1)}$$

Setting the two values of τ equal to one another, results in an equation for the controller gain.

$$\boxed{\frac{K_c K_p \theta^2}{2(K_c K_p + 1)} - \left[\frac{(\tau_p s - K_c K_p)}{2(K_c K_p + 1)}\right]^2 = 0}$$

Equation 8: Taylor Expansion Derived Kc equation

Plugging the above equation into GoalSeek results in a Kc value of 0.146 with a percent error of 19.67% and since Kc_{Taylor} is greater than Kc_{True}, the system is underdamped, as shown in figure 1.2.

A 20% increase in the controller gain from method 1 and a 20% decrease in the controller gain from method 2 were also compared to the true critical controller gain. Figure 1.3 and Table 1.1 show the comparisons. The 20% increase in method 1 resulted in a controller gain equal to the 20% decrease in controller 2.

Table 1.1: Controller Gains calculated using approximations and the percent errors

Approximation	Kc	Percent Error	Dampening
Pade	0.097	20.49	Over
Taylor	0.146	19.67	Under
20% Increase Pade	0.117	4.10	Over
20% Decrease Taylor	0.117	4.10	Over

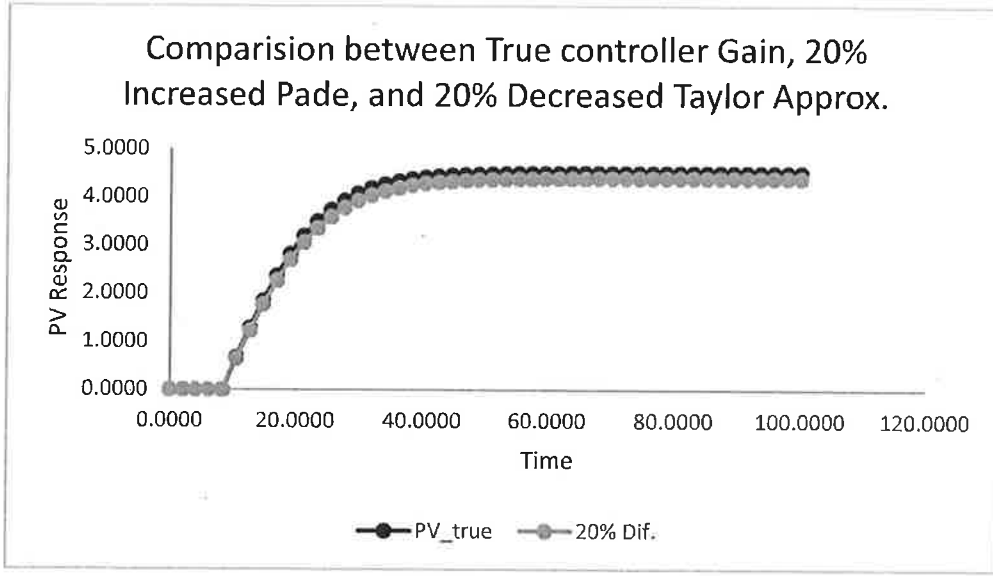

Figure 1.3: 20% difference controller gain proportional controller and critically damped controller

Problem 2

Part 1-6: Differential Time Equations for a Distillation Column

Differential time equations were derived based on the transfer functions supplied by Professor Michael Caracotsios in Midterm 2.

$$10\frac{Y_1(t)}{dt} + Y_1 = 0.1247L(t)$$

$$24\frac{Y_2(t)}{dt} + Y_2 = 0.546B(t)$$

$$16\frac{Y_3(t)}{dt} + Y_3 = -0.1246L(t)$$

$$22\frac{Y_4(t)}{dt} + Y_4 = 1.116B(t)$$

$$PV_1(t) = Y_1(t) + Y_2(t) + Y_5(t)$$

$$PV_2(t) = Y_3(t) + Y_4(t) + Y_6(t)$$

Equations 8-13: Time Differential Equations for a Distillation System

Part 7: Ruth Criterion

The characteristic equations for PV1 and PV2 are given below.

$$1 + G_{c1}G_{p11} = 0$$

$$1 + G_{c2}G_{p22} = 0$$

$$G_{c1} = 33.80\frac{(1+\tau_{I1})}{\tau_{I1}}$$

$$G_{c2} = 3.58\frac{(1+\tau_{I2})}{\tau_{I2}}$$

Solving the characteristic equations results in the following second order equations:

$$330.8\tau_{I1}s^2 + (33.2047\tau_{I1} + 320.8)s + 32.08 = 0$$

$$100.8\tau_{I2}s^2 + (5.696\tau_{I1} + 78.6)s + 1.116 = 0$$

Ruth criterion results in the following inequalities for the integral times:

$$330.8\tau_{I1} > 0; (33.2047\tau_{I1} + 320.8) > 0$$

$$\tau_{I1} > 0;\ \tau_{I1} > -9.66$$

$$\boxed{\tau_{I1} > 0}$$

$$100.8\tau_{I2} > 0;\ (5.696\tau_{I1} + 78.6) > 0$$

$$\tau_{I2} > 0;\ \tau_{I2} > -13.79$$

$$\boxed{\tau_{I2} > 0}$$

Part 8: Regulatory Control for a Distillation Column

A simulation of the controller response to disturbances was developed using Athena Visual Studios. The Athena code is a heavily modified version of the PID Regulatory Controller code written by Professor Michael Caracotsios. The code implements the above differential equations for the distillation system and can be view in Appendix I: Athena Code.

A disturbance of 0.1 more feed composition was put into the simulator. Figure 2.1 and figure 2.2 shows how the system reacts to the disturbance.

Table 2.1: Controller Settings

	Distillate Controller	Bottom Controller
Kc	33.8	3.58
τc	10	22

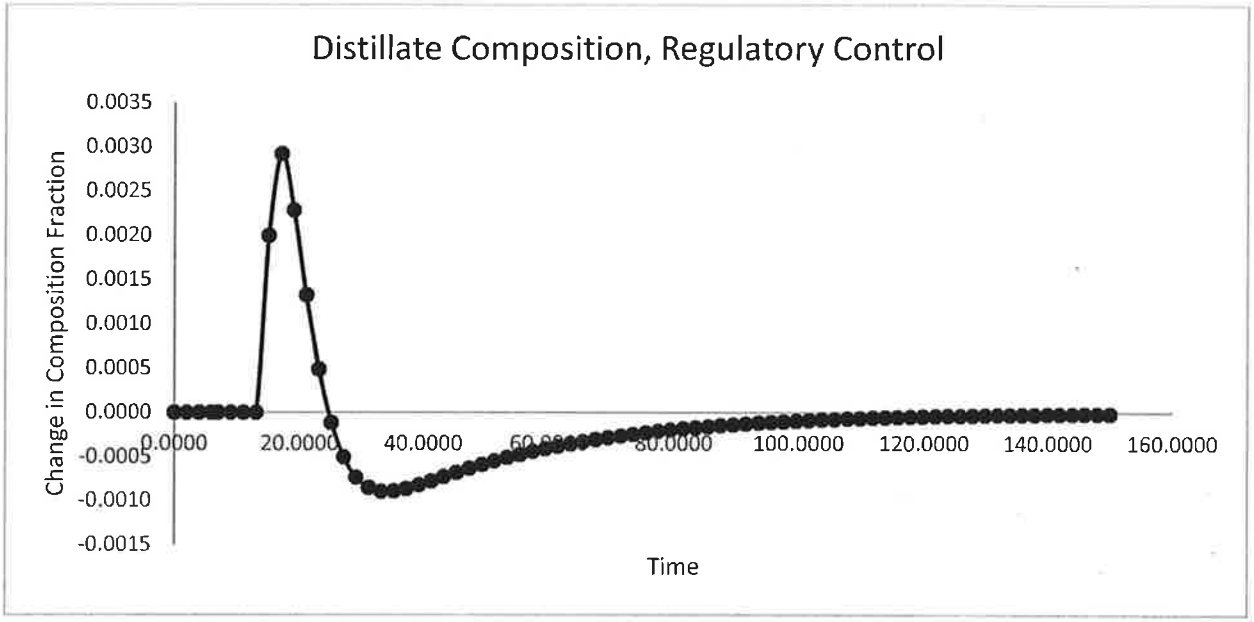

Figure 2.1: Distillate Composition Changes with a Change of 0.1 to Feed Composition

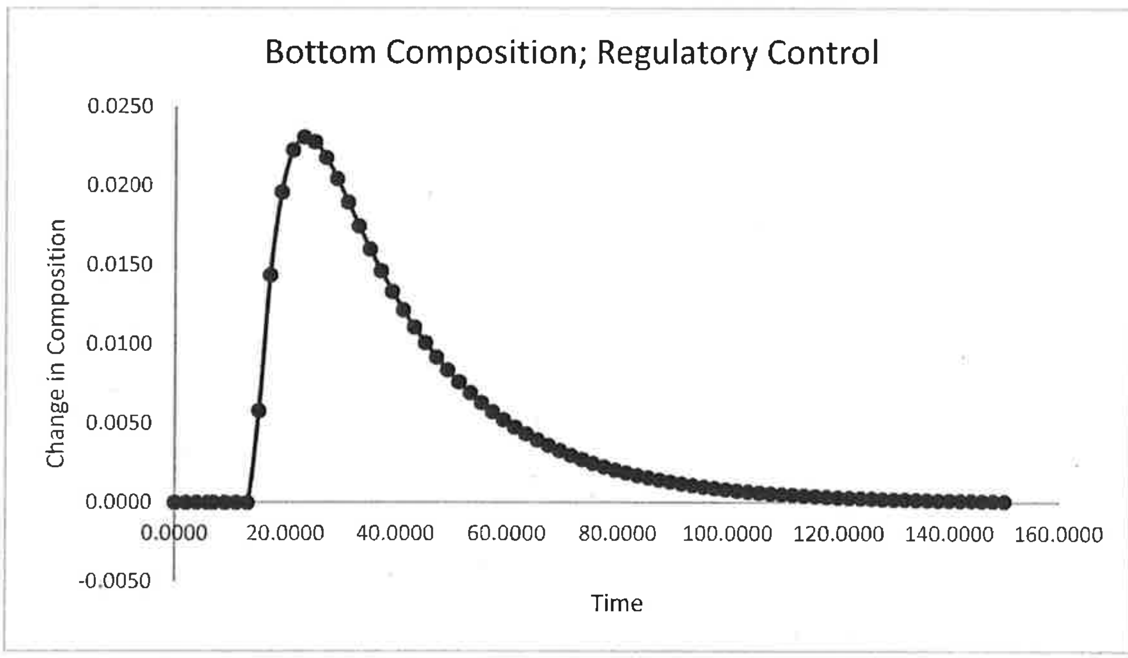

Figure 2.2: Distillate Composition Changes with a Change of 0.1 to Feed Composition

Part 9: Servo Control of Distillation Column

A step change of 0.02 in the distillate composition is made and the controller simulation results are reported below. The simulation is a modified code from the code in part 8 and can be found in Appendix I: Athena Code. The controller settings are given in the table below:

Table 2.2: Servo Controller Settings

	Distillate Controller	Bottom Controller
Kc	33.8	3.58
τc	5	15

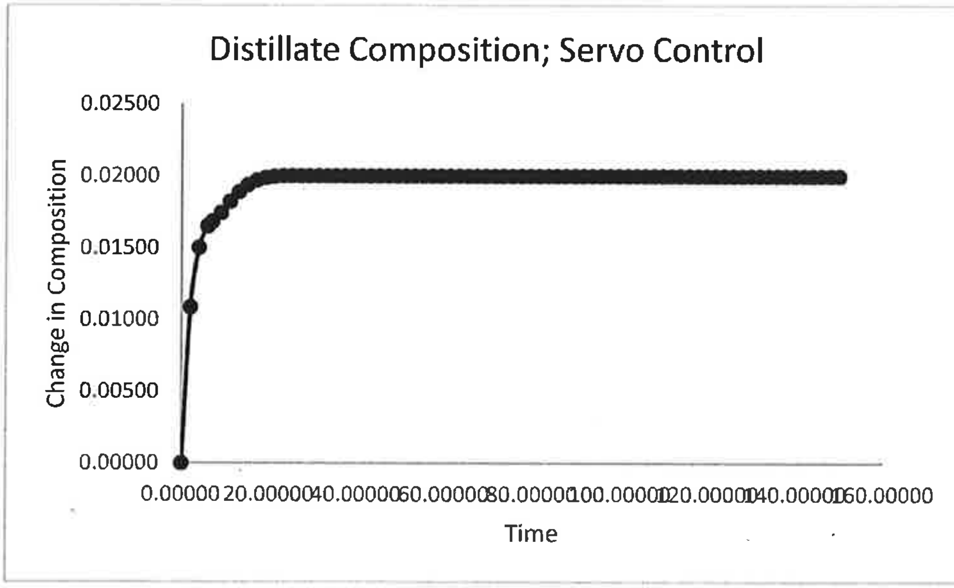

Figure 2.3: Servo Control Distillation Response

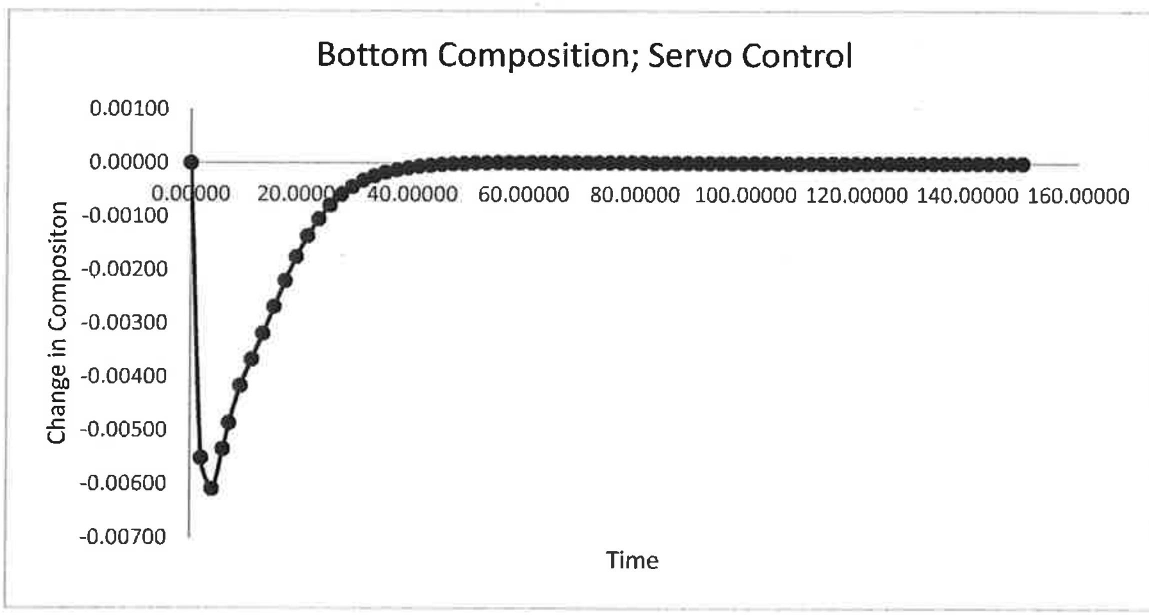

Figure 2.4: Servo Control Bottom Response

Part 10: Analyzer Dead Time Regulatory and Servo Control
An analyzer lag was given to be 5 minutes and was represented as a process time delay of 5 minutes for the system. The code used in part 10 is a modified version of the code used in part 8 and can be found in Appendix I: Athena Code.

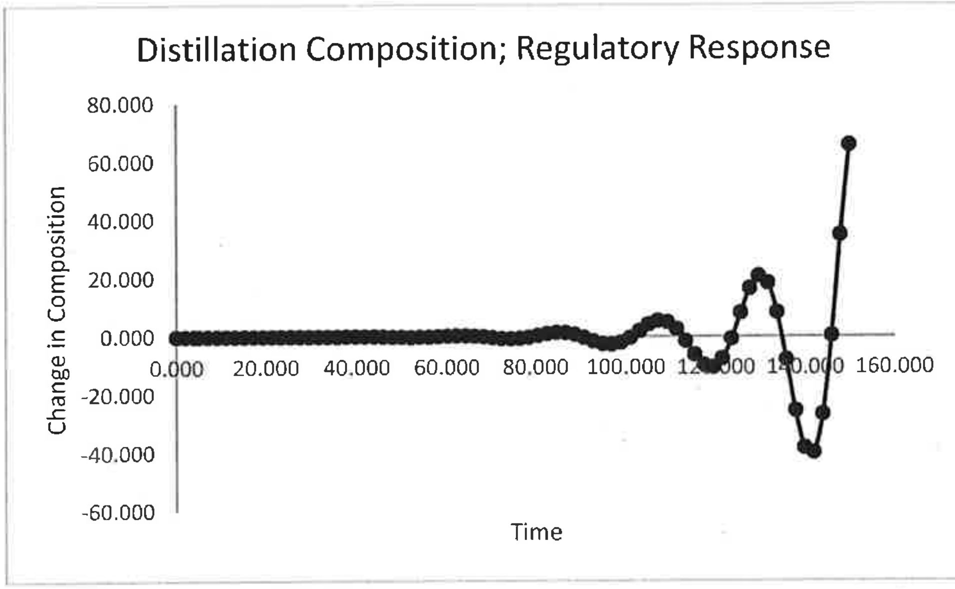

Figure 2.5: Time-Delay Regulatory Distillation Response

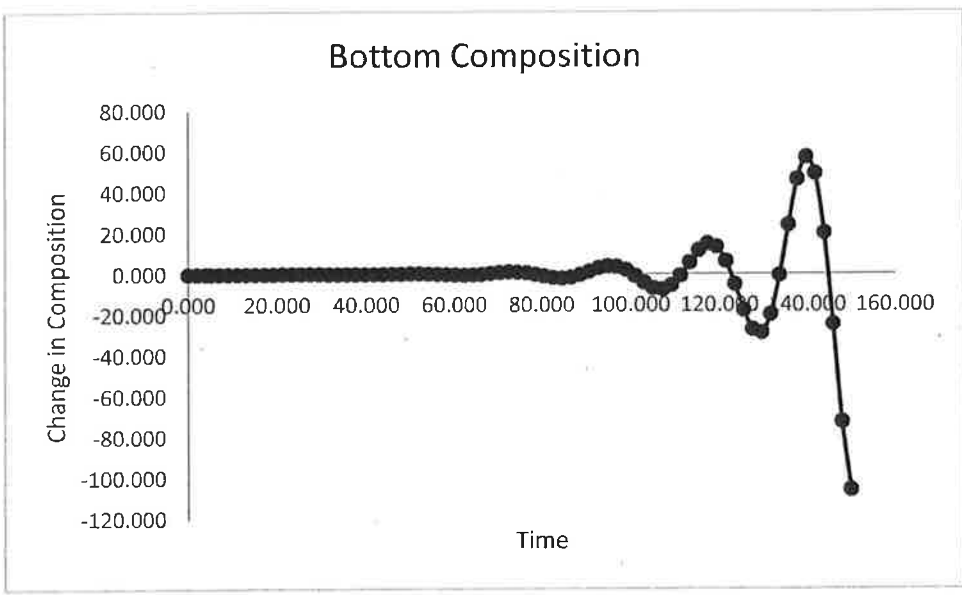

Figure 2.6: Time-Delay Regulatory Bottom Composition Response

As seen in figures 2.5 and 2.6, the settings used for the previous parts 8 and 9 no longer show stable responses now that a time-delay has been added. As such new control settings have to be explored. In the scope of this project 3 control methods have been explored: Ziegler-Nichols, Ziegler-Nichols Ultimate Gain and Cohen-Coon.

Ziegler-Nichols

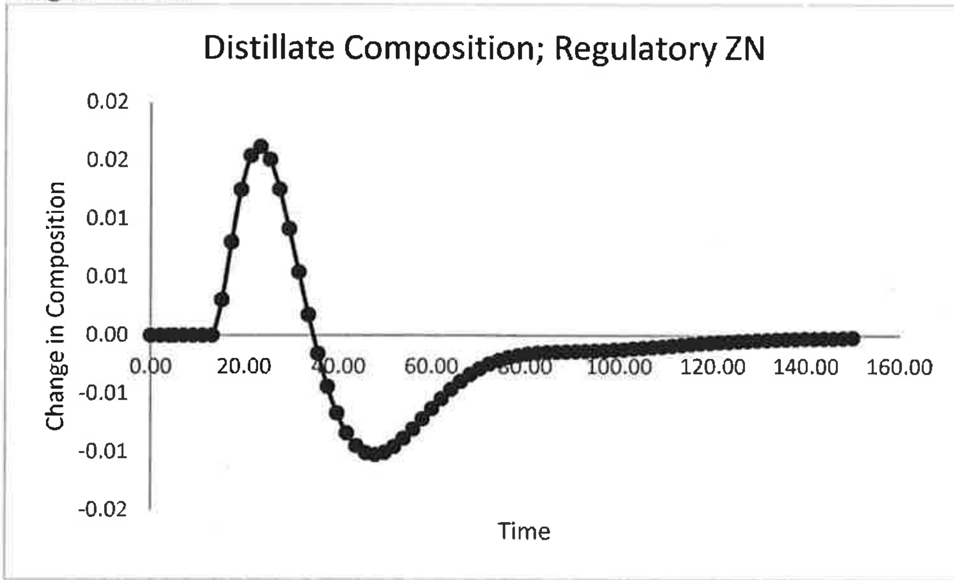

Figure 2.7: Ziegler-Nichols Regulatory Distillation Response

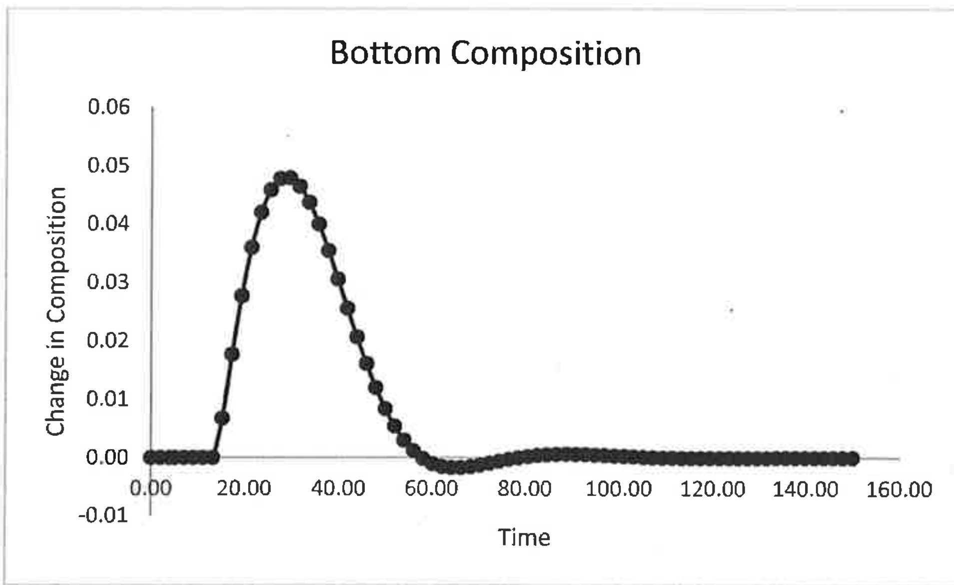

Figure 2.8: Ziegler-Nichols Regulatory Bottom Response

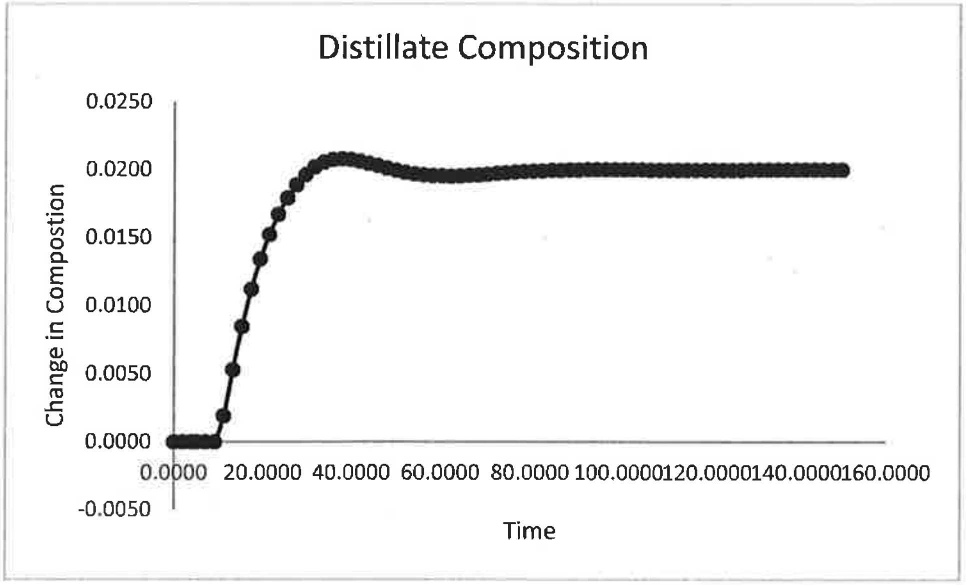

Figure 2.9: Ziegler-Nichols Servo Distillation Response

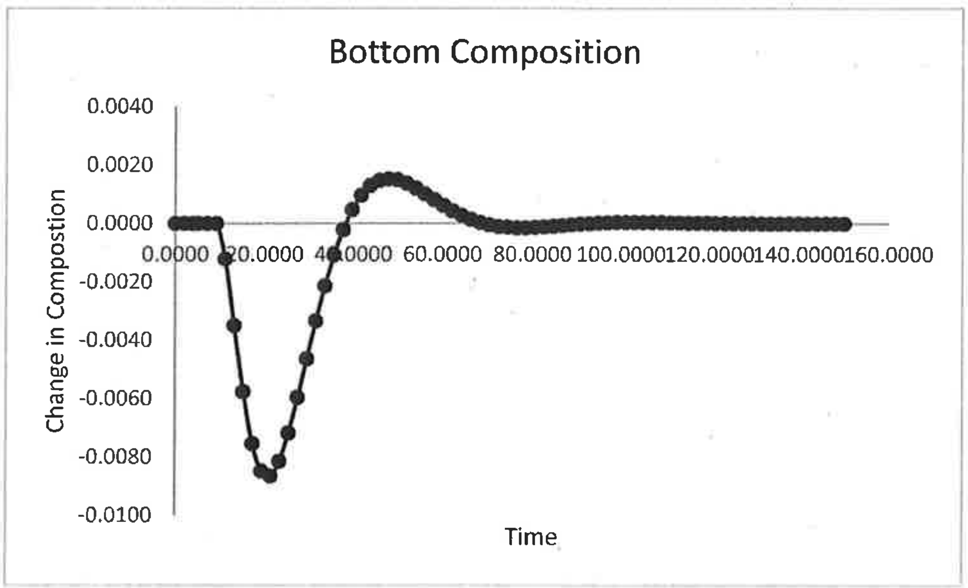

Figure 2.10: Ziegler-Nichols Servo Bottom Response

Ziegler-Nichols Ultimate Method

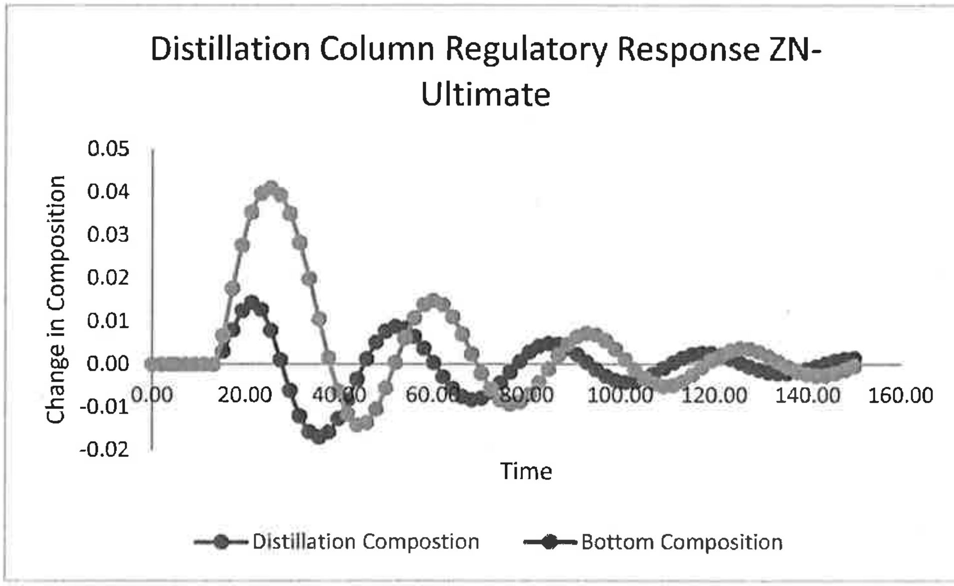

Figure 2.11: Ziegler-Nichols Ultimate Gain Regulatory Response

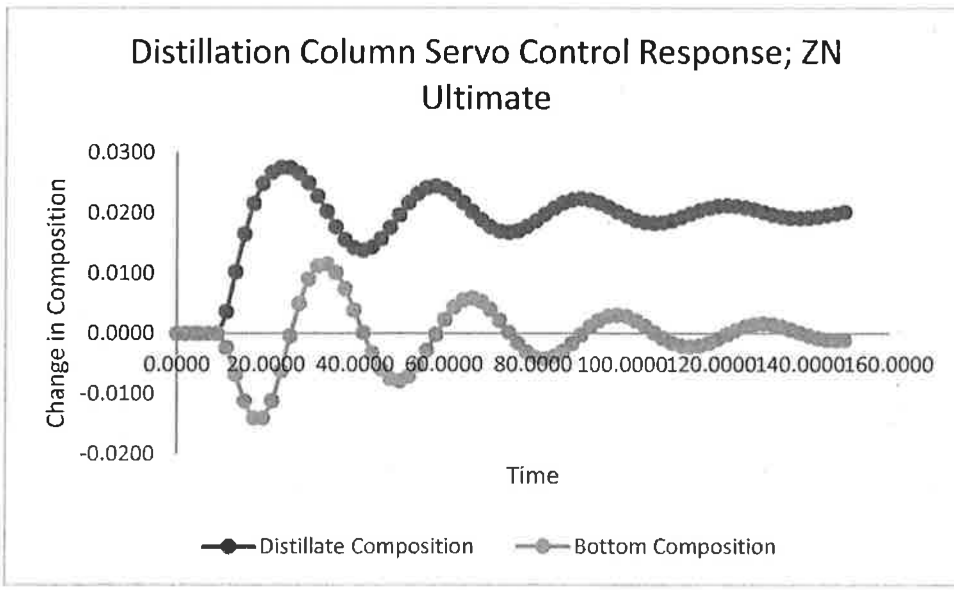

Figure 2.12: Ziegler-Nichols Ultimate Gain Servo Response

Cohen-Coon Method

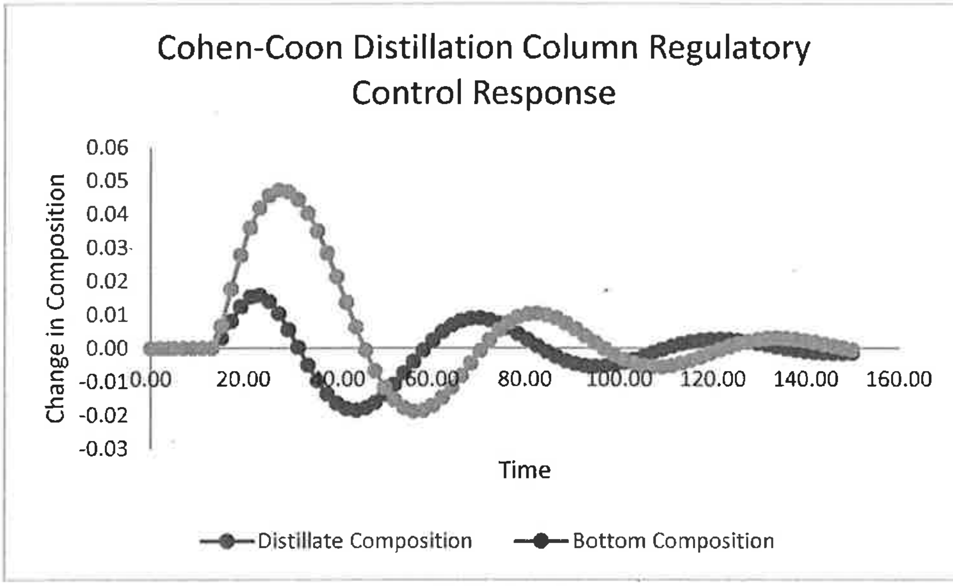

Figure 2.13: Cohen-Coon Regulatory Response

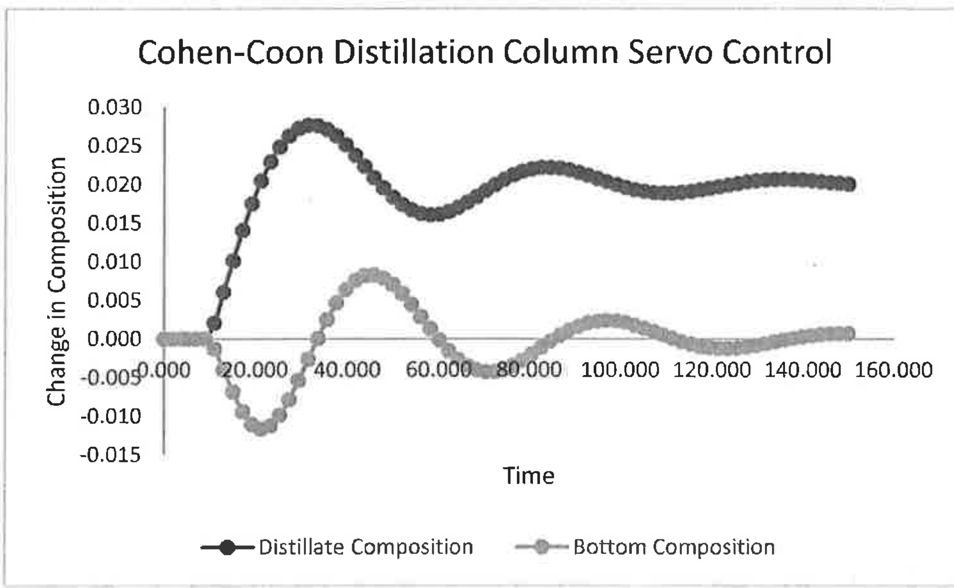

Figure 2.14: Cohen-Coon Servo Response

Based on all the responses, Ziegler-Nichols control method works the best for this distillation column. The Ziegler-Nichols control method had the fewest oscillations for both servo and regulatory response.

```
!PI Controller ; Distillation Column
!Midterm 2, Problem 8
!---------------
!Regulatory-Control Implementation in Time-Domain
!-----------------------------------------------
!Written by: Professor Michael Caracotsios; Modified by: Kevin Gassmann, Date of Modification:
4/25/17
!Contact Information(Editor): kgassm2@uic.edu
!*****************************************
 Use AthenaUtilities
 Global Taup11,Kp11,Taud1,Thetad1,Kd1,Ku1,Pu1 As Real
 Global Taup12,Kp12,Taud2,Thetad2,Kd2,Ku2,Pu2 As Real
 Global Taup21,Kp21 As Real
 Global Taup22,Kp22 As Real
 Global Kc1,Ti1,Kci1,Td1,Kcd1,SP1,DS As Real
 Global Kc2,Ti2,Kci2,Td2,Kcd2,SP2 As Real
 Global Time(1500),Y1i(1500),Y2i(1500),CO1i(1500),PV1i(1500),E1i(1500) As Real
 Global Y3i(1500),Y4i(1500),CO2i(1500),PV2i(1500),E2i(1500) As Real
 Global Y5i(1500),Y6i(1500) As Real
 Global iCount As Integer

 Taup11=10.0     !K.G. -Process Lag L(t) affecting x_D
 Kp11=0.1247     !K.G. -Process Gain L(t) affecting x_D

 Taup12=24.0    !K.G. -Process Lag B(t) affecting x_D
 Kp12=0.546     !K.G. -Process Gain B(t) affecting x_D

 Taup21=16.0     !K.G. -Process Lag L(t) affecting x_B
 Kp21=-0.1246    !K.G. -Process Gain L(t) affecting x_B

 Taup22=22.0    !K.G. -Process Lag B(t) affecting x_B
 Kp22=1.116     !K.G. -Process Gain B(t) affecting x_B

 Taud1=22.0   !K.G. -Disturbance Lag, Distillate
 Thetad1=7.0  !K.G. -Disturbance Dead time
 Kd1=0.588    !K.G. -Disturbance Gain

 Taud2=24.0   !K.G. -Disturbance Lag, Bottom
 Thetad2=7.0  !K.G. -Disturbance Dead time
 Kd2=1.41     !K.G. -Disturbance Gain

 !Ku1=0.7864   !K.G. -Ultimate Gain
 !Pu1=30.223   !K.G. -Ultimate Period

 !Ku2=0.7864   !K.G. -Ultimate Gain
 !Pu2=30.223   !K.G. -Ultimate Period

 Kc1=33.80          !K.G. -Proportional Gain for Distillate Controller
 Ti1=5.00           !K.G. -Integral Time Constant for Distillate Controller
 Kci1=Kc1/Ti1      !K.G. -Integral Gain for Distillate Controller
 Kcd1=0.0           !K.G. -Derivative Gain for Distillate Controller

 Kc2=3.58           !K.G. -Proportional Gain for Bottom Controller
 Ti2=15.4           !K.G. -Integral Time Constant for Bottom Controller
 Kci2=Kc2/Ti2      !K.G. -Integral Gain for Bottom Controller
 Kcd2=0.0           !K.G. -Derivative Gain for Bottom Controller

!Ziegler-Nichols Controller Constants
!-------------------------------------
 !If(Kci==0.0 .AND. Kcd==0.0)Then    !Proportional Controller
  !Kc=0.5*Ku
 !EndIf

 !If(Kci>0.0 .AND. Kcd==0.0)Then     !Proportional-Integral Controller
  !Kc=0.45*Ku
  !Ti=Pu/1.2
  !Kci=Kc/Ti
```

```
  !EndIf

  !If(Kci>0.0 .AND. Kcd>0.0)Then      !Proportional-Integral-Derivative Controller
   !Kc=0.60*Ku
   !Ti=Pu/2.0
   !Kci=Kc/Ti
   !Td=Pu/8.0
   !Kcd=Kc*Td
  !EndIf

  iCount=0     !K.G. -Counter for the History Profiles
  SP1=0.0      !K.G. -Distillate fraction Set Point Value
  SP2=0.0      !K.G. -Bottom fraction Set Point Value
  DS=0.1       !K.G. -Feed composition Set Point Value
```

@Initial Conditions

```
  U(1:Neq)=0.0
```

@Model Equations

```
  Dim PV1,CO1,IE1 As Real
  Dim PV2,CO2,IE2 As Real
  Dim i As Integer

 !Collect the History of the Integrated Model Variables
 !-----------------------------------------------------
  If (iRes==9)Then
   iCount=iCount+1
   Time(iCount)=T
   Y1i(iCount)=U(1)
   Y2i(iCount)=U(2)
   Y3i(iCount)=U(3)
   Y4i(iCount)=U(4)
   Y5i(iCount)=U(5)
   Y6i(iCount)=U(6)
  EndIf

 !Formulate the Feedback Control Loop Equations
 !---------------------------------------------
  Select Case(iRange)
   Case(0)
    PV1=U(1)+U(2)+U(5)
    PV2=U(3)+U(4)+U(6)

    If(iRes==0)Then
     E1i(iCount)=SP1-PV1          !K.G. -Distillate Error
     E2i(iCount)=SP2-PV2          !K.G. -Bottom Error
    EndIf
    IE1=0.0
    IE2=0.0

    Do i=1,iCount-1
     IE1=IE1+(E1i(i+1)+E1i(i))*(Time(i+1)-Time(i))/2.0
     IE2=IE2+(E2i(i+1)+E2i(i))*(Time(i+1)-Time(i))/2.0
    EndDo
    IE1=IE1+(E1i(iCount)+SP1-PV1)*(T-Time(iCount))/2.0
    CO1=Kc1*(SP1-PV1)+Kci1*IE1
    IE2=IE2+(E2i(iCount)+SP2-PV2)*(T-Time(iCount))/2.0
    CO2=Kc2*(SP2-PV2)+Kci2*IE2

    F(1)=-U(1)+Kp11*CO1
    F(2)=-U(2)+Kp12*CO2
    F(3)=-U(3)+Kp21*CO1
    F(4)=-U(4)+Kp22*CO2
    F(5)=-U(5)
    F(6)=-U(6)

   Case(1)
```

```
    PV1=U(1)+U(2)+QuadraticSpline(Time(1:iCount),Y5i(1:iCount),T-Thetad1)
    PV2=U(3)+U(4)+QuadraticSpline(Time(1:iCount),Y6i(1:iCount),T-Thetad2)

    If(iRes==9)Then
     E1i(iCount)=SP1-PV1
     E2i(iCount)=SP2-PV2
    EndIf
    IE1=0.0
    IE2=0.0

    Do i=1,iCount-1
     IE1=IE1+(E1i(i+1)+E1i(i))*(Time(i+1)-Time(i))/2.0
     IE2=IE2+(E2i(i+1)+E2i(i))*(Time(i+1)-Time(i))/2.0
    EndDo
    IE1=IE1+(E1i(iCount)+SP1-PV1)*(T-Time(iCount))/2.0
    CO1=Kc1*(SP1-PV1)+Kci1*IE1
    IE2=IE2+(E2i(iCount)+SP2-PV2)*(T-Time(iCount))/2.0
    CO2=Kc2*(SP2-PV2)+Kci2*IE2

    F(1)=-U(1)+Kp11*CO1
    F(2)=-U(2)+Kp12*CO2
    F(3)=-U(3)+Kp21*CO1
    F(4)=-U(4)+Kp22*CO2
    F(5)=-U(5)+Kd1*DS
    F(6)=-U(6)+Kd2*DS

   End Select

  !Collect History of the Process Variables and Control Output
  !------------------------------------------------------------
   If (iRes==9)Then
    PV1i(iCount)=PV1
    CO1i(iCount)=CO1
    PV2i(iCount)=PV2
    CO2i(iCount)=CO2
   EndIf
```

@Coefficient Matrix

```
  E(1)=Taup11
  E(2)=Taup12
  E(3)=Taup21
  E(4)=Taup22
  E(5)=Taud1
  E(6)=Taud2
```

@Solver Options

```
  Neq=6
  Npts=75
  Tend=150.0
  Tstop=Thetad1

  Param(28)=1                                   !K.G. -Continue Integration Past Dead Time
  Param(177)=1                                  !K.G. -Enable Integration History Access
  VariableNames=Time;Y1;Y2                  !K.G. -Variable Names
  Append=PV1;PV2;CO1;CO2                    !K.G. -Display Process Variable and Control Output
```

@After Calling Solver

```
  Dim PV1,CO1,PV2,CO2 As Real
  If(iCount==0)Then
   PV1=U(1)+U(2)+U(5)
   CO1=Kc1*(SP1-PV1)

   PV2=U(3)+U(4)+U(6)
   CO2=Kc2*(SP2-PV2)
  Else
   PV1=QuadraticSpline(Time(1:iCount),PV1i(1:iCount),Tini)
   CO1=QuadraticSpline(Time(1:iCount),CO2i(1:iCount),Tini)
```

```
  PV2=QuadraticSpline(Time(1:iCount),PV2i(1:iCount),Tini)
  CO2=QuadraticSpline(Time(1:iCount),CO2i(1:iCount),Tini)
 EndIf
```

```
!PI Controller ; Distillation Column
!Midterm 2, Problem 9
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 Global Taup12,Kp12,Taud2,Thetad2,Kd2,Ku2,Pu2 As Real
 Global Taup21,Kp21 As Real
 Global Taup22,Kp22 As Real
 Global Kc1,Ti1,Kci1,Td1,Kcd1,SP1,DS As Real
 Global Kc2,Ti2,Kci2,Td2,Kcd2,SP2 As Real
 Global Time(1500),Y1i(1500),Y2i(1500),CO1i(1500),PV1i(1500),E1i(1500) As Real
 Global Y3i(1500),Y4i(1500),CO2i(1500),PV2i(1500),E2i(1500) As Real
 Global Y5i(1500),Y6i(1500) As Real
 Global iCount As Integer

 Taup11=10.0     !K.G. -Process Lag L(t) affecting x_D
 Kp11=0.1247     !K.G. -Process Gain L(t) affecting x_D

 Taup12=24.0    !K.G. -Process Lag B(t) affecting x_D
 Kp12=0.546     !K.G. -Process Gain B(t) affecting x_D

 Taup21=16.0      !K.G. -Process Lag L(t) affecting x_B
 Kp21=-0.1246     !K.G. -Process Gain L(t) affecting x_B

 Taup22=22.0    !K.G. -Process Lag B(t) affecting x_B
 Kp22=1.116     !K.G. -Process Gain B(t) affecting x_B

 Taud1=22.0   !K.G. -Disturbance Lag, Distillate
 Thetad1=7.0  !K.G. -Disturbance Dead time
 Kd1=0.588    !K.G. -Disturbance Gain

 Taud2=24.0   !K.G. -Disturbance Lag, Bottom
 Thetad2=7.0  !K.G. -Disturbance Dead time
 Kd2=1.41     !K.G. -Disturbance Gain

 !Ku1=0.7864   !K.G. -Ultimate Gain
 !Pu1=30.223   !K.G. -Ultimate Period

 !Ku2=0.7864   !K.G. -Ultimate Gain
 !Pu2=30.223   !K.G. -Ultimate Period

 Kc1=33.80          !K.G. -Proportional Gain for Distillate Controller
 Ti1=5.00           !K.G. -Integral Time Constant for Distillate Controller
 Kci1=Kc1/Ti1      !K.G. -Integral Gain for Distillate Controller
 Kcd1=0.0           !K.G. -Derivative Gain for Distillate Controller

 Kc2=3.58           !K.G. -Proportional Gain for Bottom Controller
 Ti2=15.0           !K.G. -Integral Time Constant for Bottom Controller
 Kci2=Kc2/Ti2      !K.G. -Integral Gain for Bottom Controller
 Kcd2=0.0           !K.G. -Derivative Gain for Bottom Controller

!Ziegler-Nichols Controller Constants
!-----------------------------------
 !If(Kci==0.0 .AND. Kcd==0.0)Then    !Proportional Controller
  !Kc=0.5*Ku
 !EndIf

 !If(Kci>0.0 .AND. Kcd==0.0)Then     !Proportional-Integral Controller
  !Kc=0.45*Ku
  !Ti=Pu/1.2
  !Kci=Kc/Ti
```

```
  !EndIf

  !If(Kci>0.0 .AND. Kcd>0.0)Then      !Proportional-Integral-Derivative Controller
   !Kc=0.60*Ku
   !Ti=Pu/2.0
   !Kci=Kc/Ti
   !Td=Pu/8.0
   !Kcd=Kc*Td
  !EndIf

  iCount=0     !K.G. -Counter for the History Profiles
  SP1=0.02      !K.G. -Distillate fraction Set Point Value
  SP2=0.0      !K.G. -Bottom fraction Set Point Value
  DS=0.0       !K.G. -Feed composition Set Point Value
```

@Initial Conditions

```
  U(1:Neq)=0.0
```

@Model Equations

```
  Dim PV1,CO1,IE1 As Real
  Dim PV2,CO2,IE2 As Real
  Dim i As Integer

 !Collect the History of the Integrated Model Variables
 !-------------------------------------------------------
  If (iRes==9)Then
   iCount=iCount+1
   Time(iCount)=T
   Y1i(iCount)=U(1)
   Y2i(iCount)=U(2)
   Y3i(iCount)=U(3)
   Y4i(iCount)=U(4)
   Y5i(iCount)=U(5)
   Y6i(iCount)=U(6)
  EndIf

 !Formulate the Feedback Control Loop Equations
 !---------------------------------------------
  Select Case(iRange)
   Case(0)
    PV1=U(1)+U(2)+U(5)
    PV2=U(3)+U(4)+U(6)

    If(iRes==0)Then
     E1i(iCount)=SP1-PV1           !K.G. -Distillate Error
     E2i(iCount)=SP2-PV2           !K.G. -Bottom Error
    EndIf
    IE1=0.0
    IE2=0.0

    Do i=1,iCount-1
     IE1=IE1+(E1i(i+1)+E1i(i))*(Time(i+1)-Time(i))/2.0
     IE2=IE2+(E2i(i+1)+E2i(i))*(Time(i+1)-Time(i))/2.0
    EndDo
    IE1=IE1+(E1i(iCount)+SP1-PV1)*(T-Time(iCount))/2.0
    CO1=Kc1*(SP1-PV1)+Kci1*IE1
    IE2=IE2+(E2i(iCount)+SP2-PV2)*(T-Time(iCount))/2.0
    CO2=Kc2*(SP2-PV2)+Kci2*IE2

    F(1)=-U(1)+Kp11*CO1
    F(2)=-U(2)+Kp12*CO2
    F(3)=-U(3)+Kp21*CO1
    F(4)=-U(4)+Kp22*CO2
    F(5)=-U(5)
    F(6)=-U(6)

   Case(1)
```

```
    PV1=U(1)+U(2)+QuadraticSpline(Time(1:iCount),Y5i(1:iCount),T-Thetad1)
    PV2=U(3)+U(4)+QuadraticSpline(Time(1:iCount),Y6i(1:iCount),T-Thetad2)

    If(iRes==9)Then
     E1i(iCount)=SP1-PV1
     E2i(iCount)=SP2-PV2
    EndIf
    IE1=0.0
    IE2=0.0

    Do i=1,iCount-1
     IE1=IE1+(E1i(i+1)+E1i(i))*(Time(i+1)-Time(i))/2.0
     IE2=IE2+(E2i(i+1)+E2i(i))*(Time(i+1)-Time(i))/2.0
    EndDo
    IE1=IE1+(E1i(iCount)+SP1-PV1)*(T-Time(iCount))/2.0
    CO1=Kc1*(SP1-PV1)+Kci1*IE1
    IE2=IE2+(E2i(iCount)+SP2-PV2)*(T-Time(iCount))/2.0
    CO2=Kc2*(SP2-PV2)+Kci2*IE2

    F(1)=-U(1)+Kp11*CO1
    F(2)=-U(2)+Kp12*CO2
    F(3)=-U(3)+Kp21*CO1
    F(4)=-U(4)+Kp22*CO2
    F(5)=-U(5)+Kd1*DS
    F(6)=-U(6)+Kd2*DS

   End Select

 !Collect History of the Process Variables and Control Output
 !------------------------------------------------------------
  If (iRes==9)Then
   PV1i(iCount)=PV1
   CO1i(iCount)=CO1
   PV2i(iCount)=PV2
   CO2i(iCount)=CO2
  EndIf

@Coefficient Matrix
  E(1)=Taup11
  E(2)=Taup12
  E(3)=Taup21
  E(4)=Taup22
  E(5)=Taud1
  E(6)=Taud2

@Solver Options
  Neq=6
  Npts=75
  Tend=150.0
  Tstop=Thetad1

  Param(28)=1                                 !K.G. -Continue Integration Past Dead Time
  Param(177)=1                                !K.G. -Enable Integration History Access
  VariableNames=Time;Y1;Y2                !K.G. -Variable Names
  Append=PV1;PV2;CO1;CO2                  !K.G. -Display Process Variable and Control Output

@After Calling Solver
  Dim PV1,CO1,PV2,CO2 As Real
  If(iCount==0)Then
   PV1=U(1)+U(2)+U(5)
   CO1=Kc1*(SP1-PV1)

   PV2=U(3)+U(4)+U(6)
   CO2=Kc2*(SP2-PV2)
  Else
   PV1=QuadraticSpline(Time(1:iCount),PV1i(1:iCount),Tini)
   CO1=QuadraticSpline(Time(1:iCount),CO2i(1:iCount),Tini)
```

```
 PV2=QuadraticSpline(Time(1:iCount),PV2i(1:iCount),Tini)
 CO2=QuadraticSpline(Time(1:iCount),CO2i(1:iCount),Tini)
EndIf
```

```
!PI Controller ; Distillation Column
!Midterm 2, Problem 10
!---------------
!Regulatory-Control Implementation in Time-Domain
!----------------------------------------------------
!Written by: Professor Michael Caracotsios; Modified by: Kevin Gassmann, Date of Modification:
4/25/17
!Contact Information(Editor): kgassm2@uic.edu
!*****************************************
 Use AthenaUtilities
 Global Taup11,Kp11,Taud1,Thetad1,Kd1,Ku1,Pu1 As Real
 Global Taup12,Kp12,Taud2,Thetad2,Kd2,Ku2,Pu2 As Real
 Global Taup21,Kp21 As Real
 Global Taup22,Kp22, ThetaP As Real
 Global Kc1,Ti1,Kci1,Td1,Kcd1,SP1,DS As Real
 Global Kc2,Ti2,Kci2,Td2,Kcd2,SP2,SM As Real
 Global Time(1500),Y1i(1500),Y2i(1500),CO1i(1500),PV1i(1500),E1i(1500) As Real
 Global Y3i(1500),Y4i(1500),CO2i(1500),PV2i(1500),E2i(1500) As Real
 Global Y5i(1500),Y6i(1500) As Real
 Global iCount As Integer

 ThetaP=5.0       !K.G. -Analyzer Deadtime
 Taup11=10.0      !K.G. -Process Lag L(t) affecting x_D
 Kp11=0.1247      !K.G. -Process Gain L(t) affecting x_D

 Taup12=24.0     !K.G. -Process Lag B(t) affecting x_D
 Kp12=0.546      !K.G. -Process Gain B(t) affecting x_D

 Taup21=16.0       !K.G. -Process Lag L(t) affecting x_B
 Kp21=-0.1246      !K.G. -Process Gain L(t) affecting x_B

 Taup22=22.0     !K.G. -Process Lag B(t) affecting x_B
 Kp22=1.116      !K.G. -Process Gain B(t) affecting x_B

 Taud1=22.0   !K.G. -Disturbance Lag, Distillate
 Thetad1=7.0  !K.G. -Disturbance Dead time
 Kd1=0.588    !K.G. -Disturbance Gain

 Taud2=24.0   !K.G. -Disturbance Lag, Bottom
 Thetad2=7.0  !K.G. -Disturbance Dead time
 Kd2=1.41     !K.G. -Disturbance Gain

 Ku1=30.52844   !K.G. -Ultimate Gain
 Pu1=17.10552   !K.G. -Ultimate Period

 Ku2=6.775555   !K.G. -Ultimate Gain
 Pu2=18.44271   !K.G. -Ultimate Period

 Kc1=0.00             !K.G. -Proportional Gain for Distillate Controller
 Ti1=0.00             !K.G. -Integral Time Constant for Distillate Controller
 !Kci1=Kc1/Ti1        !K.G. -Integral Gain for Distillate Controller
 Kcd1=0.0             !K.G. -Derivative Gain for Distillate Controller

 Kc2=0.00             !K.G. -Proportional Gain for Bottom Controller
 Ti2=0.00             !K.G. -Integral Time Constant for Bottom Controller
 !Kci2=Kc2/Ti2        !K.G. -Integral Gain for Bottom Controller
 Kcd2=0.0             !K.G. -Derivative Gain for Bottom Controller

 SM = 2.0            !K.G. -Stability Margin for controller tuning
!Ziegler-Nichols Controller Constants
!-------------------------------------
 !Kc1=0.9/(SM*Kp11)*(Taup11/ThetaP)
 !Ti1=3.33*ThetaP
 !Kc2=0.9/(SM*Kp22)*(Taup22/ThetaP)
 !Ti2=3.33*ThetaP
 !Kci1=Kc1/Ti1
 !Kci2=Kc2/Ti2
```

```
!Cohen-Coon Controller Constants
!-----------------------------------
 !Kc1=0.9/(SM*Kp11)*((Taup11/ThetaP)+0.092)
 !Ti1=3.33*ThetaP*((Taup11+0.082*ThetaP)/(Taup11+2.22*ThetaP))
 !Kc2=0.9/(SM*Kp22)*((Taup22/ThetaP)+0.092)
 !Ti2=3.33*ThetaP*((Taup22+0.082*ThetaP)/(Taup22+2.22*ThetaP))
 !Kci1=Kc1/Ti1
 !Kci2=Kc2/Ti2

!Ziegler-Nichols Ultimate Gain Controller Constants
!------------------------------------
  Kc1=0.45*Ku1            !K.G. -Proportional Integral Controller
  Ti1=Pu1/1.2
  Kci1=Kc1/Ti1

  Kc2=0.45*Ku2
  Ti2=Pu2/1.2
  Kci2=Kc2/Ti2


  iCount=0     !K.G. -Counter for the History Profiles
  SP1=0.02      !K.G. -Distillate fraction Set Point Value
  SP2=0.0      !K.G. -Bottom fraction Set Point Value
  DS=0.0       !K.G. -Feed composition Set Point Value
```

@Initial Conditions

```
  U(1:Neq)=0.0
```

@Model Equations

```
  Dim PV1,CO1,IE1 As Real
  Dim PV2,CO2,IE2 As Real
  Dim i As Integer

 !Collect the History of the Integrated Model Variables
 !-----------------------------------------------------
  If (iRes==9)Then
   iCount=iCount+1
   Time(iCount)=T
   Y1i(iCount)=U(1)
   Y2i(iCount)=U(2)
   Y3i(iCount)=U(3)
   Y4i(iCount)=U(4)
   Y5i(iCount)=U(5)
   Y6i(iCount)=U(6)
  EndIf

 !Formulate the Feedback Control Loop Equations
 !---------------------------------------------
  Select Case(iRange)
   Case(0)
    PV1=U(1)+U(2)+U(5)
    PV2=U(3)+U(4)+U(6)

    If(iRes==0)Then
     E1i(iCount)=SP1-PV1            !K.G. -Distillate Error
     E2i(iCount)=SP2-PV2            !K.G. -Bottom Error
    EndIf
    IE1=0.0
    IE2=0.0

    Do i=1,iCount-1
     IE1=IE1+(E1i(i+1)+E1i(i))*(Time(i+1)-Time(i))/2.0
     IE2=IE2+(E2i(i+1)+E2i(i))*(Time(i+1)-Time(i))/2.0
    EndDo
    IE1=IE1+(E1i(iCount)+SP1-PV1)*(T-Time(iCount))/2.0
```

```
     CO1=Kc1*(SP1-PV1)+Kci1*IE1
     IE2=IE2+(E2i(iCount)+SP2-PV2)*(T-Time(iCount))/2.0
     CO2=Kc2*(SP2-PV2)+Kci2*IE2

     F(1)=-U(1)
     F(2)=-U(2)
     F(3)=-U(3)
     F(4)=-U(4)
     F(5)=-U(5)
     F(6)=-U(6)

    Case(1)
     PV1=QuadraticSpline(Time(1:iCount),Y1i(1:iCount),T-
ThetaP)+QuadraticSpline(Time(1:iCount),Y2i(1:iCount),T-ThetaP)
     PV2=QuadraticSpline(Time(1:iCount),Y3i(1:iCount),T-
ThetaP)+QuadraticSpline(Time(1:iCount),Y4i(1:iCount),T-ThetaP)

     If(iRes==9)Then
      E1i(iCount)=SP1-PV1
      E2i(iCount)=SP2-PV2
     EndIf
     IE1=0.0
     IE2=0.0

     Do i=1,iCount-1
      IE1=IE1+(E1i(i+1)+E1i(i))*(Time(i+1)-Time(i))/2.0
      IE2=IE2+(E2i(i+1)+E2i(i))*(Time(i+1)-Time(i))/2.0
     EndDo
     IE1=IE1+(E1i(iCount)+SP1-PV1)*(T-Time(iCount))/2.0
     CO1=Kc1*(SP1-PV1)+Kci1*IE1
     IE2=IE2+(E2i(iCount)+SP2-PV2)*(T-Time(iCount))/2.0
     CO2=Kc2*(SP2-PV2)+Kci2*IE2

     F(1)=-U(1)+Kp11*CO1
     F(2)=-U(2)+Kp12*CO2
     F(3)=-U(3)+Kp21*CO1
     F(4)=-U(4)+Kp22*CO2
     F(5)=-U(5)
     F(6)=-U(6)
     Case(2)
     PV1=QuadraticSpline(Time(1:iCount),Y1i(1:iCount),T-
ThetaP)+QuadraticSpline(Time(1:iCount),Y2i(1:iCount),T-ThetaP)
     PV1=PV1+QuadraticSpline(Time(1:iCount),Y5i(1:iCount),T-Thetad1)
     PV2=QuadraticSpline(Time(1:iCount),Y3i(1:iCount),T-
ThetaP)+QuadraticSpline(Time(1:iCount),Y4i(1:iCount),T-ThetaP)
     PV2=PV2+QuadraticSpline(Time(1:iCount),Y6i(1:iCount),T-Thetad2)
     If(iRes==9)Then
      E1i(iCount)=SP1-PV1
      E2i(iCount)=SP2-PV2
     EndIf
     IE1=0.0
     IE2=0.0

     Do i=1,iCount-1
      IE1=IE1+(E1i(i+1)+E1i(i))*(Time(i+1)-Time(i))/2.0
      IE2=IE2+(E2i(i+1)+E2i(i))*(Time(i+1)-Time(i))/2.0
     EndDo
     IE1=IE1+(E1i(iCount)+SP1-PV1)*(T-Time(iCount))/2.0
     CO1=Kc1*(SP1-PV1)+Kci1*IE1
     IE2=IE2+(E2i(iCount)+SP2-PV2)*(T-Time(iCount))/2.0
     CO2=Kc2*(SP2-PV2)+Kci2*IE2

     F(1)=-U(1)+Kp11*CO1
     F(2)=-U(2)+Kp12*CO2
     F(3)=-U(3)+Kp21*CO1
     F(4)=-U(4)+Kp22*CO2
     F(5)=-U(5)+Kd1*DS
```

```
    F(6)=-U(6)+Kd2*DS

   End Select

  !Collect History of the Process Variables and Control Output
  !-----------------------------------------------------------
   If (iRes==9)Then
    PV1i(iCount)=PV1
    CO1i(iCount)=CO1
    PV2i(iCount)=PV2
    CO2i(iCount)=CO2
   EndIf
```

@Coefficient Matrix

```
  E(1)=Taup11
  E(2)=Taup12
  E(3)=Taup21
  E(4)=Taup22
  E(5)=Taud1
  E(6)=Taud2
```

@Solver Options

```
  Neq=6
  Npts=75
  Tend=150.0
  Tstop=ThetaP;Thetad1

  Param(28)=1                                       !K.G. -Continue Integration Past Dead Time
  Param(177)=1                                      !K.G. -Enable Integration History Access
  VariableNames=Time;Y1;Y2                      !K.G. -Variable Names
  Append=PV1;PV2;CO1;CO2                        !K.G. -Display Process Variable and Control Output
```

@After Calling Solver

```
  Dim PV1,CO1,PV2,CO2 As Real
  If(iCount==0)Then
   PV1=U(1)+U(2)+U(5)
   CO1=Kc1*(SP1-PV1)

   PV2=U(3)+U(4)+U(6)
   CO2=Kc2*(SP2-PV2)
  Else
   PV1=QuadraticSpline(Time(1:iCount),PV1i(1:iCount),Tini)
   CO1=QuadraticSpline(Time(1:iCount),CO2i(1:iCount),Tini)

   PV2=QuadraticSpline(Time(1:iCount),PV2i(1:iCount),Tini)
   CO2=QuadraticSpline(Time(1:iCount),CO2i(1:iCount),Tini)
  EndIf
```